

### 7.3 Classwork: Logarithm and Graphing Challenge — Markscheme

1. [Maximum mark: 9]

(a) Solve  $\log_2 x + \log_4 x + \log_8 x = 11$ .

Change all terms to base 2:  $\log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{1}{2} \log_2 x$  and  $\log_8 x = \frac{1}{3} \log_2 x$  **M1**

$$\log_2 x + \frac{1}{2} \log_2 x + \frac{1}{3} \log_2 x = 11 \quad \textbf{A1}$$

$$\frac{11}{6} \log_2 x = 11 \Rightarrow \log_2 x = 6 \quad \textbf{A1}$$

$$x = 2^6 = 64 \quad \textbf{A1}$$

(b) Show that  $\log_a b \cdot \log_b a = 1$ .

Using change of base,  $\log_a b = \frac{\ln b}{\ln a}$  and  $\log_b a = \frac{\ln a}{\ln b}$ . **M1**

$$\log_a b \cdot \log_b a = \frac{\ln b}{\ln a} \cdot \frac{\ln a}{\ln b} = 1 \quad \textbf{A1 AG}$$

(c) Given  $\log_a b = 4$ , find  $\log_{a^2} b^3$ .

$$\log_{a^2} b^3 = \frac{\log_a b^3}{\log_a a^2} = \frac{3 \log_a b}{2} \quad \textbf{M1 A1}$$

$$= \frac{3}{2} \cdot 4 = 6 \quad \textbf{A1}$$

2. [Maximum mark: 5]

Solve  $(\log_3 x)^2 - 5\log_3 x + 6 = 0$ .

**Solution:**

Let  $u = \log_3 x$ , so  $u^2 - 5u + 6 = 0$ .

**M1**

$(u - 2)(u - 3) = 0$

**A1**

$u = 2 \quad \text{or} \quad u = 3$

**A1**

Back-substitute:  $\log_3 x = 2 \Rightarrow x = 3^2 = 9$

**A1**

$\log_3 x = 3 \Rightarrow x = 3^3 = 27$

**A1**

(Both solutions are valid since each gives  $x > 0$ .)

3. [Maximum mark: 9]

Let  $f(x) = 3^{x-1} + 2$ .

(a) Find  $f^{-1}(x)$ .

Set  $y = 3^{x-1} + 2$  and solve for  $x$ :

$$y - 2 = 3^{x-1} \quad \text{M1}$$

$$\log_3(y - 2) = x - 1 \quad \text{A1}$$

$$x = 1 + \log_3(y - 2) \quad \text{A1}$$

$$f^{-1}(x) = 1 + \log_3(x - 2) \quad \text{A1}$$

(b) Sketch of  $y = f(x)$ ,  $y = f^{-1}(x)$ , and  $y = x$ :

- Correct sketch of  $f$  (increasing exponential through  $(1, 3)$ , rising to upper-right) **A1**
- Horizontal asymptote  $y = 2$  drawn and labelled **A1**
- Correct sketch of  $f^{-1}$  (reflection of  $f$  across  $y = x$ ; passes through  $(3, 1)$ ) **A1**
- Vertical asymptote  $x = 2$  drawn and labelled **A1**
- Line  $y = x$  drawn **A1**

*Reference points used to verify the sketches:*  $f(0) = \frac{1}{3} + 2 \approx 2.33$ ,  $f(1) = 3$ ,  $f(2) = 5$ ; correspondingly  $f^{-1}(2.33) \approx 0$ ,  $f^{-1}(3) = 1$ ,  $f^{-1}(5) = 2$ .

4. [Maximum mark: 12]

(a) Solve  $2^x = 20$ .

Take  $\log_2$  of both sides:

**M1**

$$x = \log_2 20$$

**A1**

*Equivalent forms accepted:*  $\frac{\ln 20}{\ln 2}$ ,  $\frac{\log 20}{\log 2}$ ,  $2 + \log_2 5$ .

(b) Solve  $5^{x-1} = 12$ .

Take  $\log_5$  of both sides:

**M1**

$$x - 1 = \log_5 12$$

**A1**

$$x = 1 + \log_5 12$$

**A1**

*Equivalent form:*  $x = 1 + \frac{\ln 12}{\ln 5}$ .

(c)  $P(t) = 50 \cdot 2^t$ ; find  $t$  such that  $P = 1500$ .

$$50 \cdot 2^t = 1500 \Rightarrow 2^t = 30$$

**M1 A1**

$$t = \log_2 30$$

**A1**

*Equivalent forms:*  $\frac{\ln 30}{\ln 2}$ ,  $1 + \log_2 15$ .

(d) Solve  $5^{2x} = 3^{x+1}$ .

Take  $\ln$  of both sides:

**M1**

$$2x \ln 5 = (x + 1) \ln 3$$

**A1**

Collect  $x$ -terms:  $2x \ln 5 - x \ln 3 = \ln 3$ , so  $x(2 \ln 5 - \ln 3) = \ln 3$

**M1**

$$x = \frac{\ln 3}{2 \ln 5 - \ln 3} = \frac{\ln 3}{\ln(25/3)}$$

**A1**